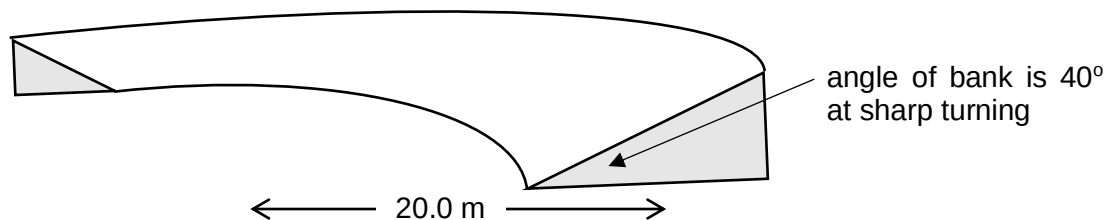


$$\text{Difference} = 0.0077 \text{ rad s}^{-1}$$

- 11** The section of a cycling track where sharper turns are made has a horizontal radius of 20.0 m, and is banked at 40° to the horizontal.



Given that the maximum speed of the cyclist of mass 80 kg is 60 km h^{-1} , what is the difference in frictional force that the bicycle tyres must provide compared to an unbanked track?

- A** 140 N **B** 450 N **C** 760 N **D** 1200 N

Ans: C

For a non-banked track, centripetal force is provided by the frictional force

$$f_1 = mv^2 / r = (80)(60 \times 1000/3600)^2 / 20.0 = 1110 \text{ N}$$

For a 40° banked track,

$$N \cos 40^\circ = mg + f_2 \sin 40^\circ$$

$$N \sin 40^\circ + f_2 \cos 40^\circ = mv^2 / r$$

$$f_2 = 347 \text{ N}$$

$$\text{Difference} = 1110 - 347 = 763 \text{ N} \approx 760 \text{ N}.$$

- 12** A planet has an average density 4050 kg m^{-3} . Given that the acceleration of free fall near the